

Supporting information for: Temperature-Dependent Implicit-Solvent Model of Polyethylene Glycol in Aqueous Solution

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1 MARTINI Force Field

In our preliminary investigation, the MARTINI based CG model^{S1} of PEG, originally parametrized at 296 K, was tested at different temperatures. A single 36-mer chain was placed into the simulation box containing 1950 MARTINI water beads and 4 MARTINI anti-freeze particles. After equilibration, a 400 ns trajectory was obtained using the leap-frog (md) integrator with a timestep of 8 fs. Simulations were performed at constant pressure 1 bar but various temperatures (294 K, 320 K and 371 K), controlled with the Berendsen barostat and the velocity-rescale thermostat, respectively. As recommended by the MARTINI scheme, Lennard-Jones interaction forces were smoothly switched to zero in a distance from 0.9 nm to 1.2 nm. The radius of gyration, R_g , of the 36-mer chain was evaluated for each simulation run. Contrary to the atomistic simulations and our final CG model, the polymer coil rather swells with increasing temperature as shown in Figure S1.

2 Iterative Boltzmann Inversion (IBI)

To derive the effective nonbonded pair potential, IBI was employed in an aqueous solution of PEG oligomers. The longer oligomer was used, the shallower the attraction well in the potential (Figure S2). Though we chose two different target radial distribution functions (RDFs), in the case of the trimer (between terminal units and between middle units, respectively), the potentials developed to almost identical structures in both cases. The only noticeable and expected difference is for the desolvation barrier. Thus the RDF between middle units was chosen as the target RDF when IBI of longer oligomers (i.e., nonamer) was employed.

As shown in Figure S3, a rather short cutoff distance of 1.2 nm in the IBI procedure is sufficient.

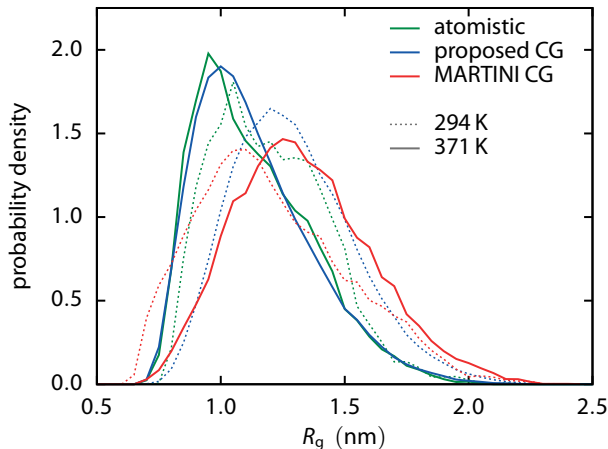


Figure S1: Distributions of the radius of gyration, R_g , of PEG 36-mer at 294 K (dashed lines) and 371 K (solid lines) employing atomistic REMD simulations, the MARTINI CG model^{S1} and the CG model proposed in this work. The MARTINI CG model, contrary to our model, predicts swelling of the coil as the temperature increases. However, it shall be noted that the MARTINI CG model was parametrized only at 296 K and validity at higher temperatures (e.g., 320 K) was not assumed.

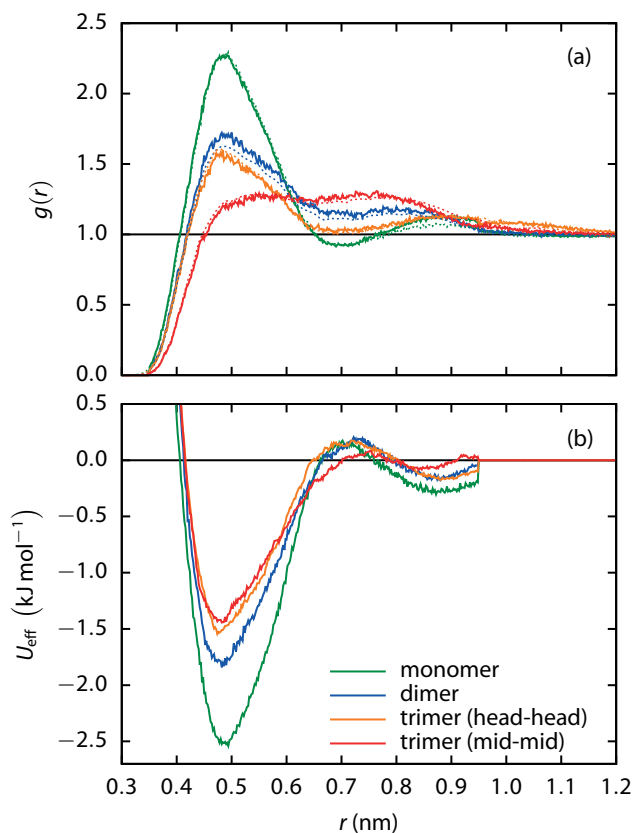


Figure S2: The radial distribution functions (a) and the converged effective pair nonbonded potentials (b) from preliminary tests of IBI in solutions of monomer, dimer and trimer at 356 K. The radial distribution functions between monomer molecules, EO units of dimer, terminal EO units of trimer and middle EO units of trimer from the atomistic simulations (panel (a), dotted lines) were chosen as the target RDFs. Though there is a clear difference in the target RDFs for differently chosen units in the trimer, the obtained final CG potentials are rather similar. Note that the abrupt cutoff of the nonbonded potential at 0.95 nm as shown here was not used in the production simulations, where a smoothly decaying function was applied instead.

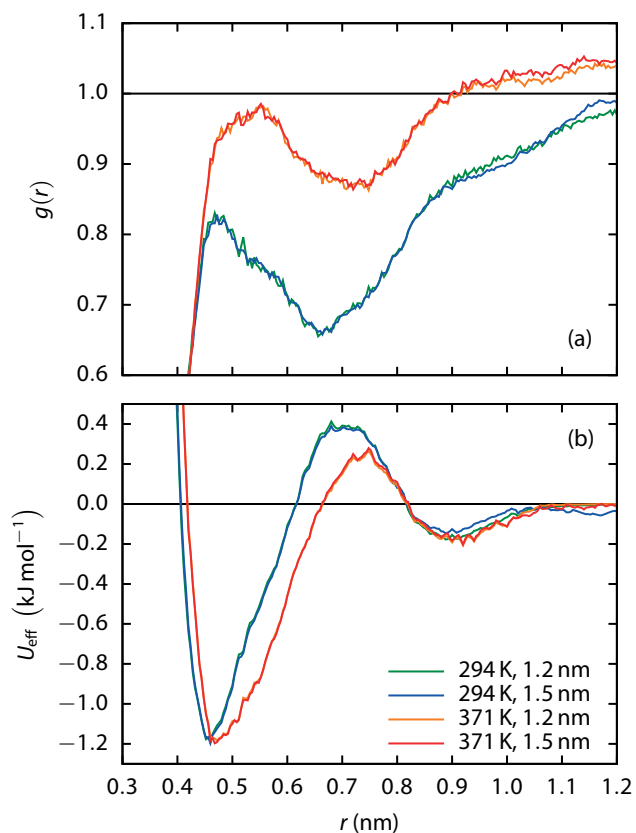


Figure S3: The effect of an imposed cutoff distance on the CG nonbonded potential during the IBI procedure. The converged effective pair nonbonded potentials U_{eff} (b) and the corresponding RDFs (a), obtained in ≈ 1 M aqueous solution of PEG nonamer are shown. In the first case, the cutoff distance was set to 1.5 nm with damping starting at 1.3 nm. In the other case, a shorter distance 1.2 nm with damping from 1.0 nm was chosen.

3 Top-down Postrefinement

When the nonbonded potentials emerged from the IBI of monomers or trimers are applied to a PEG polymer, they turn out to be too attractive. In the consequence, the polymer chain collapses into a dense globule even at room temperature (Figure S4), contrary to the good solvent regime observed in experiments. The nonbonded potential derived from the nonamers partly enables swollen states though a further refinement is needed to achieve an agreement with the experimental data.

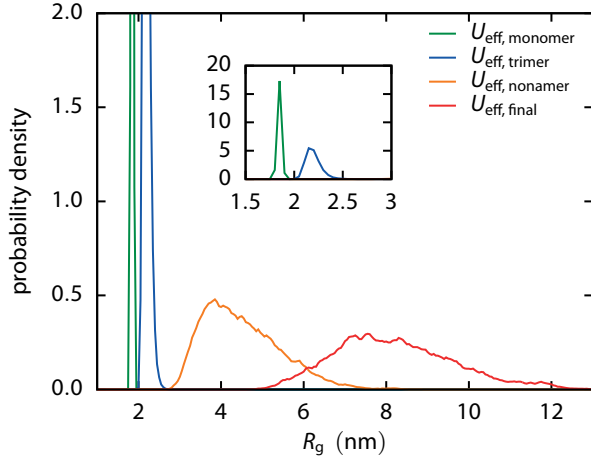


Figure S4: Distributions of the radius of gyration R_g of a PEG 795-mer at 294 K when various effective nonbonded pair potentials are applied. The potentials derived using IBI in solutions of PEG monomer and trimer bring the polymer into a densely collapsed globule. Even the potential derived from a solution of nonamer is more attractive when compared to the experimental data. The inset shows in full the narrow distributions of the dense globules.

The analytic expression, eq 1 can be used to fit the nonbonded potential, derived with IBI in a solution of nonamers, up to the desolvation barrier. The resulting set of parameter values, in a temperature range from 270 K to 450 K, is summarized in Table S1.

$$U_{\text{nonbonded}}(r) = \underbrace{\left(\frac{n}{n-m}\right) \left(\frac{n}{m}\right)^{m/(n-m)} \varepsilon \left[\left(\frac{\sigma}{r}\right)^n - \left(\frac{\sigma}{r}\right)^m \right]}_{\text{Mie potential}} + \underbrace{\gamma e^{-\left(\frac{r-\mu}{\delta}\right)^2}}_{\text{Gaussian potential}}, \quad (1)$$

As mentioned before, our choice of the analytic expression does not describe the relatively shallow second minimum in the nonbonded potential. However, omitting this minimum turns the nonbonded potential from too attractive to too repulsive (Figure S5), which may be compensated by increas-

Table S1: Final sets of parameters for the pair nonbonded potential (eq 1) after fixing the parameters $n = 8.00$, $\gamma = 0.4841 \text{ kJ mol}^{-1}$, $\delta = 0.1064 \text{ nm}$ and with ε adjusted from $1.193 \text{ kJ mol}^{-1}$ to $1.372 \text{ kJ mol}^{-1}$.

T (K)	m	σ (nm)	μ (nm)
270	10.74	0.4039	0.6818
294	10.14	0.4081	0.6888
320	8.63	0.4120	0.6955
347	7.50	0.4158	0.7018
371	6.53	0.4188	0.7070
396	5.94	0.4238	0.7154
422	5.24	0.4277	0.7219
450	5.09	0.4312	0.7277

ing the (also temperature-independent) value of ε . It shall be mentioned that the unaltered nonbonded potential does not follow the empirical and theoretical scaling law $R_g(N) = lN^\nu$ while the modified form does (Figure S6). The effect of various ε values on R_g is demonstrated in Figure S7.

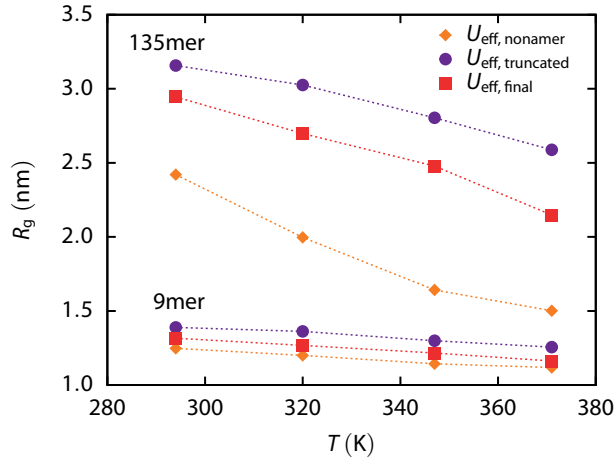


Figure S5: Radius of gyration R_g of CG PEG 9-mer and 135-mer as a function of temperature when various effective non-bonded pair potentials are applied. Comparing to the experimental data (not shown), the full potential derived directly with IBI in a solution of PEG nonamer, $U_{\text{eff, nonamer}}$ is more attractive, while the same potential truncated after desolvation barrier, $U_{\text{eff, trunc}}$ is less attractive.

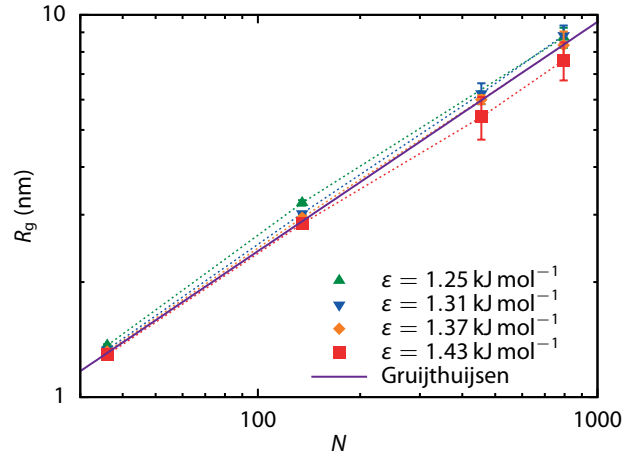


Figure S7: Radius of gyration R_g at 294 K as a function of the chain length at various ϵ values used in the nonbonded potential (eq 1) during the parameter refinement. The experimental data^{S2,S3} are shown for comparison.

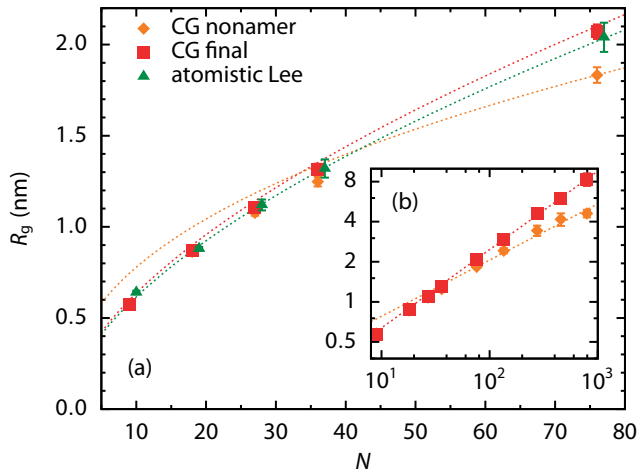


Figure S6: Radius of gyration R_g at 294 K as a function of the chain length when the nonbonded potentials derived with IBI of nonamer is applied in the CG model. Comparing to the adjusted final nonbonded potential and experimental data, the simulation yields more collapsed conformations and the radius of gyration does not follow the scaling $R_g(N) = lN^\nu$ (dashed lines).

4 Structural Comparison

We performed an atomistic REMD of the PEG 36-mer and compared chain structural properties to the CG model. The simulation box contained a single polymer chain and 5824 TIP3P water molecules; the simulation setup is described in detail in the main text.

First, bonded parameters, such as distributions of the bond length, the bond angle and the dihedral angle were scrutinized. Virtual sites placed at the centers of mass of the intended CG beads were used as reference points when evaluating the atomistic simulations. The CG model retains much of bonded structural details as shown in Figure S8. Though the dihedral angle distribution is more prominent in the atomistic simulations near room temperature, the proposed CG model exhibits almost the identical distribution when compared to the CG model of Lee et al.^{S1}

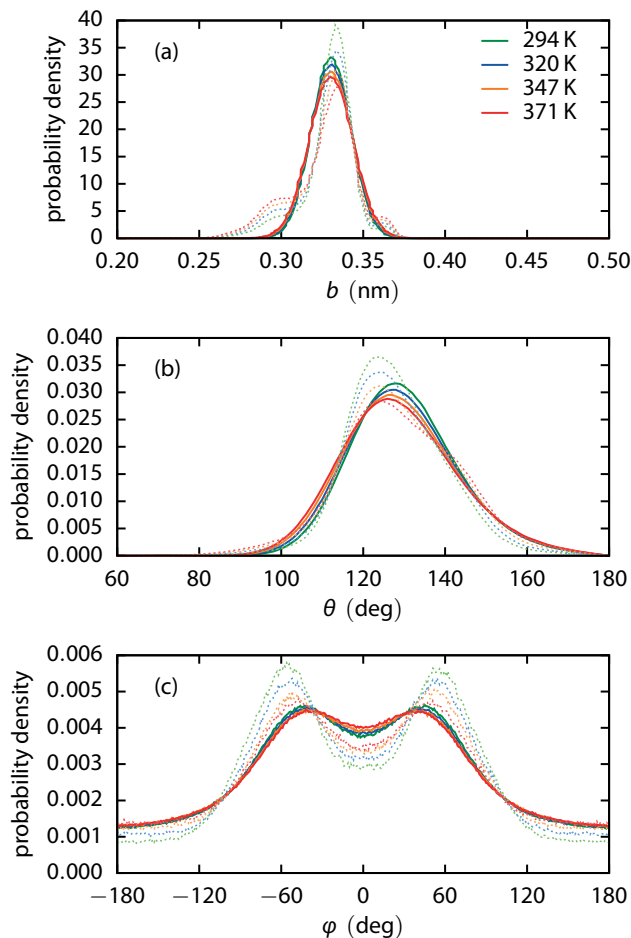


Figure S8: Comparison between the atomistic simulation and the CG model of PEG 36-mer at various temperatures. Distributions of (a) the bond length b , (b) the bond angle θ , and (c) the dihedral angle φ are shown. The data originating from the virtual sites in the atomistic simulations are shown with dashed lines, the actual distributions in the CG model with solid lines.

The radius of gyration R_g and the end-to-end distance R_{ee} were used as representative chain properties in Figure S9.

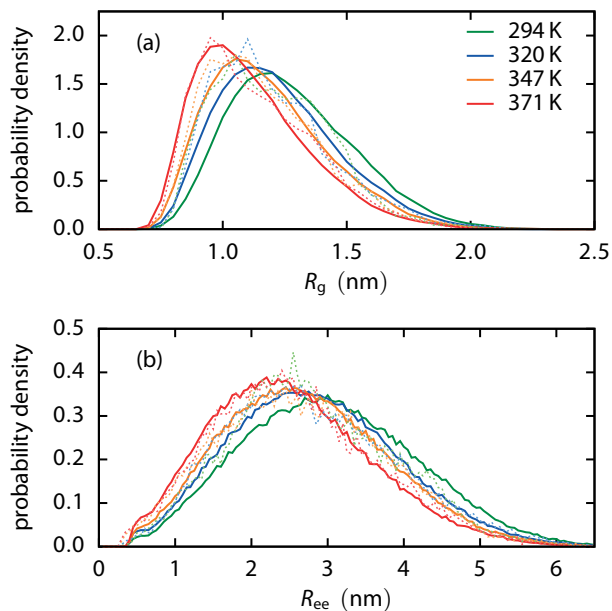


Figure S9: Distributions of (a) the radius of gyration R_g and (b) the end-to-end distance R_{ee} of the PEG 36-mer at various temperatures. The data originating from the virtual sites in the atomistic simulations are shown with dashed lines, the actual distributions in the CG model with solid lines.

5 Single-chain Simulation Trajectories

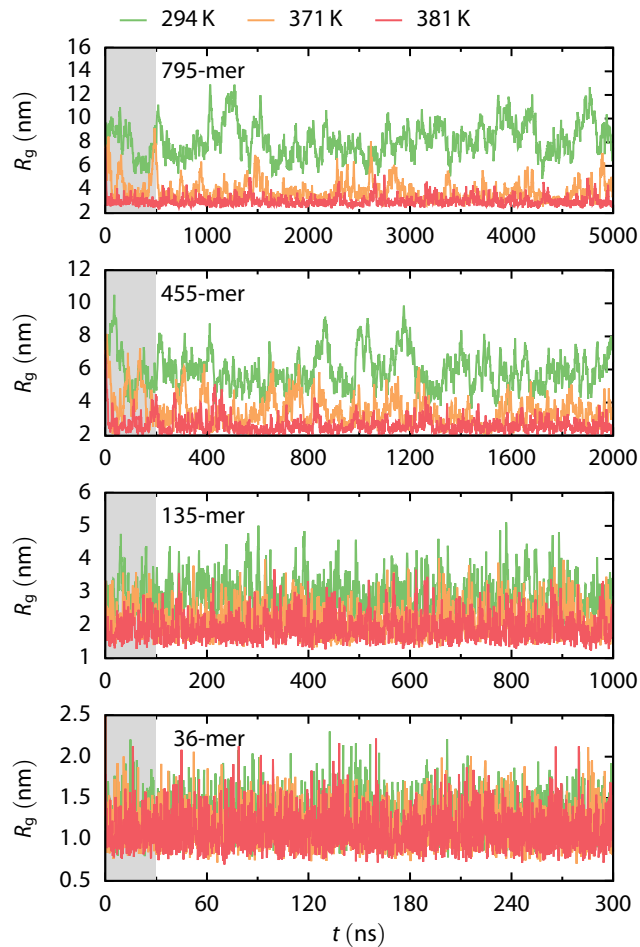


Figure S10: Simulation trajectories of the radius of gyration of polymer chains. For each polymer chain, trajectories in a good solvent regime (294 K), around LCST (371 K) and in a bad solvent regime (381 K) are shown. The first tenth of the trajectory (gray background) is considered as an equilibration phase and not included in the analysis.

6 The Second Virial Coefficient

The following relation is used to convert between the concentration (number per unit volume) based second virial coefficient B_2 and the mass density (mass per unit volume) based second virial coefficient A_2 :^{S4}

$$B_2 = \frac{A_2 M^2}{N_A}, \quad (2)$$

where M is the molar mass of the polymer chain (kg mol^{-1}) and N_A the Avogadro constant.

The molar mass M of a polymer chain being N monomers long is calculated as

$$M = NM_m, \quad M_m = 44.05 \text{ g mol}^{-1}. \quad (3)$$

7 Kirkwood-Buff Integrals

The Kirkwood-Buff Integrals (KBIs) were evaluated for three oligomers (9-, 18-, and 27-mer) at 294 K as a function of concentration. Values were compared to experimental data.^{S5} While the longer oligomers achieve both qualitative and quantitative agreement, the nonamer does not exhibit experimentally observed properties. For the nonamer, however, the effects of end groups are significant.^{S6} In our model, the PEG chains are terminated with methyl caps, while the experiments were conducted with an unmodified PEG, having hydrophilic hydroxyl groups at the ends. That is why the proposed model does not provide matching results for the very short oligomers.

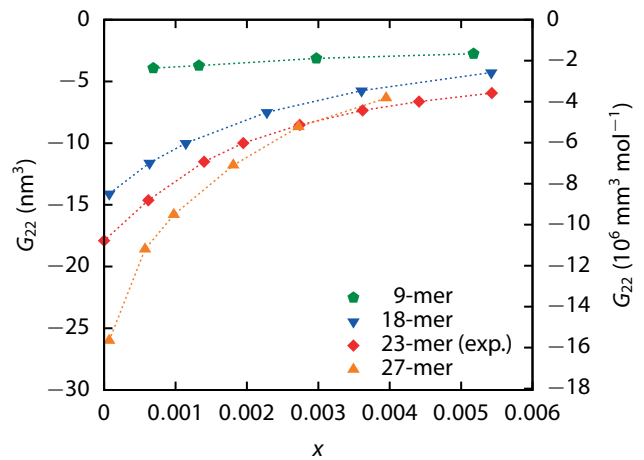


Figure S11: The Kirkwood-Buff integral G_{22} between chains for three oligomers (see the legend) in a good solvent regime (294 K) employing the proposed CG model. Experimental data^{S5} for 23-mer ($M = 1000 \text{ g mol}^{-1}$) are shown for comparison. KBI values are plotted as a function of concentration, expressed by the molar fraction x of the oligomer. Note that the right-hand-side y-axis allows an easy comparison with Figures 7–8 in the referenced paper.^{S5}

References

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